ESOF 322

Software Engineering

LECTURE: 2

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OUTLINE:

- History of GIT
- GIT
- Review of Navigating the Command Line and Bash
- Git Settings
- etc

HISTORY

of GIT

1. CVS – CONCURRENT VERSIONS SYSTEM (C. 1990–2008)

One of the first widely used version control systems.

Characteristics:

- Managed versions of individual files, not entire projects.
- Centralized model – a single main repository.

Limitations:

- Poor branching support.
- No strong data integrity.
- Slow operations.
- Used in early open-source projects, including Linux in its early stages.

2. BITKEEPER (2000–2018)

A **commercial**, distributed version control system known for speed.
From 2002, it was **free for the Linux community** under a special license.

Why it mattered:

- Linus Torvalds used it to manage the Linux kernel codebase.

Crisis in 2005:

- Andrew Tridgell created a tool called SourcePuller by reverse-engineering BitKeeper's protocol.
- BitMover (Larry McVoy's company) considered this a license violation and revoked free access.

2. BITKEEPER (2000–2018)

What happened next:

- In 2016, BitKeeper was open-sourced under the Apache 2.0 license.
- Last release: 2018.
- Today, it’s practically abandoned—Git completely replaced it.

3. THE BIRTH OF GIT (2005)

After losing BitKeeper, Linus Torvalds set requirements for a new tool:

- **Free and open source.**
- **Distributed** (every user has the full history).
- **Extremely fast** (apply a patch in <3 seconds).
- **Resilient to corruption.**

Timeline:

- April 3, 2005 – development begins.
- June 2005 – first release.

3. THE BIRTH OF GIT (2005)

The name ***Git***:

- *Global Information Tracker* (when it works well).
- *Goddamn Idiotic Truckload of ***__ (when it doesn't).

  Officially: *the stupid content tracker.*

GIT - GLOBAL INFORMATION TRACKER

- Git is a free and open source distributed version control system designed to handle everything from small to very large projects with speed and efficiency.
- Git is easy to learn and has a tiny footprint with lightning fast performance. It outclasses SCM tools like Subversion, CVS, Perforce, and ClearCase with features like cheap local branching, convenient staging areas, and multiple workflows.

Docs: https://git-scm.com/doc

WHAT IS GIT? (SIMPLE DEFINITION FOR BEGINNERS)

Git is a version control system. Its main job is to track every change in your code and allow you to go back to previous versions if needed. It also lets you compare changes between versions.

But **Git** does more than that:

1. It allows multiple developers to work on the same project at the same time by creating alternative versions of the code, called branches.
2. These branches can later be merged back together. If two people changed the same part of the code, Git will show a conflict that needs to be resolved.
3. This makes Git perfect for collaborative work and for projects that evolve quickly.

GIT IN AGILE & PLAN-DRIVEN

- In Agile development, where code changes often and quality can drop over time, Git helps by making refactoring and merging much easier.
- In more rigid, plan-driven approaches, branching and merging might be less frequent, but Git is still useful for tracking history and avoiding mistakes.

WINDOWS VS LINUX

Command Conventions

WINDOWS VS LINUX COMMAND CONVENTIONS

- In Windows command-line environments such as Command Prompt (cmd) (and the more modern PowerShell), you typically run programs by typing their name, for example:

```
ping
```

- This actually runs ping.exe (the operating system automatically appends the .exe or .com extension). After the command, you add parameters separated by spaces.
- In Windows, **parameters are usually preceded by / (slash) or sometimes - (hyphen/dash).**

WINDOWS VS LINUX COMMAND CONVENTIONS

The traditional way to get help for a command is by using /? or -h or /h. For example:

`ping /?` , `dir /?`

This displays the available options for that command.

```
C:\>ping /?
Usage: ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-v TOS]
            [-r count] [-s count] [[-j host-list] | [-k host-list]]
            [-w timeout] [-R] [-S srcaddr] [-c compartment] [-p]
            [-4] [-6] target_name
Options:
    -t             Ping the specified host until stopped.
                   To see statistics and continue - type Control-Break;
                   To stop - type Control-C.
    -a             Resolve addresses to hostnames.
    -n count       Number of echo requests to send.
```

WINDOWS VS LINUX COMMAND CONVENTIONS

In Linux/Unix systems, a different convention is used:

- Parameters are preceded by - **(single dash)** for short options (usually one letter), and these can often be combined.
- Parameters are preceded by -- (**double dash)** for long, descriptive options, which cannot be combined.

```
student@pi5v:~ $ ping --help
Usage
  ping [options] <destination>
Options:
  <destination>      dns name or ip address
  -a                 use audible ping
  -A                 use adaptive ping
  -B                 sticky source address
  -c <count>         stop after <count> replies
```

WHY GIT WORKS

on Multiple Consoles in Windows

Since Git was originally developed to manage the Linux kernel and is open source, the Windows version of Git can run in at least three different command-line environments:

- **Git Bash**: A Linux-like shell that recognizes common Linux commands such as ls, touch, mkdir, etc.
- **Command Prompt (cmd)**: The classic Windows command line, compatible with MS-DOS conventions.
- **PowerShell**: A powerful Windows shell, significantly different from cmd. It supports some Linux-like commands, but there are important differences between PowerShell and Bash, which often confuse beginners when using Git in PowerShell.

WHY GIT WORKS

on Multiple Consoles in Windows

Regardless of which console you use, Git follows the Linux/Unix convention for command-line options:

- A single dash (-) for short, single-letter options (which can often be combined).
- A double dash (--) for long, descriptive options (which cannot be combined).

Console	Typical Use	Linux Commands Supported?
`Git Bash`	Yes	Fully supported
`CMD`	Yes	No
`PowerShell`	Yes	Partially (aliases)

GIT SETTINGS

Let’s start by configuring Git. Every user—whether on Windows or Linux—has their own local Git settings. Here are the key commands:

```
git config --global user.name "Jacob Pach"       # Sets your name for commits
git config --global user.email "jpach@mtech.edu" # Sets your email for commits
git config --global core.editor "code --wait"    # Sets VS Code as the default editor
git config --global -e                           # Opens the global config file for editing
git config --global core.autocrlf true           # Handles line endings (important on Windows)
```

GIT CONFIG

--global core.autocrlf true

This Git configuration command ensures consistent handling of line endings across different operating systems. When set to true, Git automatically converts Windows-style carriage return + line feed (CRLF) to Unix-style line feed (LF) when committing, and converts back to CRLF when checking out files on Windows. This helps prevent issues caused by inconsistent line endings in collaborative projects involving multiple platforms.

`CRLF <--> LF`

GIT COMMANDS

- init
- status
- add
- commit
- log
- branch
- switch / checkout
- merge

GIT INIT

This command initializes a new Git repository in the current directory. It creates a hidden .git folder that stores all version control data. After running git init, you can start tracking changes, committing files, and using other Git features. It's typically the first step when starting a new project with Git.

```
.git
+---hooks
+---info
+---logs
+---objects
+---refs
|   COMMIT_EDITMSG
|   config
|   description
|   HEAD
|   index
```

GIT STATUS

This command displays the current state of the working directory and staging area. It shows which files have been modified, staged for commit, or are untracked. It helps developers understand what changes are pending and whether they need to add, commit, or discard changes. It's a key tool for tracking progress and avoiding mistakes during version control.

```
$ git status
On branch master

No commits yet

nothing to commit (create/copy files and use "git add" to track)
```

GIT `STATUS -S` - SHORT STATUS VIEW

This command shows a condensed version of git status, making it easier to quickly scan changes. It uses symbols to indicate the state of each file:

- ?? – Untracked file (not added to Git yet)
- A – File added to staging area
- M – Modified file
- D – Deleted file
- R – Renamed file

Each line shows the status and the filename, making it ideal for fast reviews during development. `git status -s` is especially useful when working with many files. It helps you stay focused and avoid clutter from long status messages.

GIT WORKFLOW: UNDERSTANDING THE THREE MAIN AREAS

- **Directory / Working Space**

  This is where you actively edit files. Changes made here are not yet tracked by Git unless added to the staging area. It reflects your current work-in-progress.

- **Staging Area**

  Also known as the index (“checkpoint”), this is a preparation zone where you place changes that you intend to commit. It allows you to selectively group changes before saving them to the repository.

- **Repository**

  This is the permanent storage area where committed changes are saved. It contains the full history of your project and allows you to track, revert, or collaborate on code over time.

ADDING A FILE TO OUR DIRECTORY

Let's try adding a file to our directory by creating the first file, `firstFile.txt`, using the touch command. Then, we can check the status of our repository to see how Git tracks this new file.

```
$ touch firstFile.txt
$ git status
On branch master
No commits yet
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        firstFile.txt
nothing added to commit but untracked files present (use "git add" to track)
```

ADDING A FILE TO OUR DIRECTORY

```
$ touch firstFile.txt
$ git status
On branch master
No commits yet
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        firstFile.txt
nothing added to commit but untracked files present (use "git add" to track)
```

Git has detected a new file, `firsFile.txt`, but it is untracked, meaning it is not yet being monitored by Git, and suggests using `git add <file>` to start tracking the file and include it in the next commit. This is a typical state right after creating a new file in a freshly initialized repository.


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THANK

You